ZHR at StanceEval-2026: Surface-Matched Synthetic Demonstrations and Metric-Aware Decoding for Arabic Cross-Target Stance Detection

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Abstract

We describe the ZHR system for both tracks of StanceEval-2026, Arabic stance detection on Mawqif-XT, where Track 1 evaluates on seen targets and Track 2 on unseen ones. The system combines fine-tuned Arabic encoders, a few-shot classifier that retrieves labelled examples for a served instruction model, and generative models adapted with LoRA, and turns their blended probabilities into labels with a threshold rule that abstains to NONE unless one stance class is probable enough, with thresholds fixed on the development split. We report the whole sequence of experiments, including eight that lowered F_{avg2} and were not adopted. The main finding is that transfer to a new target is carried by the demonstrations, the labelled examples retrieved into the prompt, rather than by the model itself. Generated on-target tweets in the example pool gave the largest gain on both tracks, matching the surface form of the test tweets helped where the target has a distinctive style, and doubling the examples hurt. Our official results are F_{avg2} 0.896 with accuracy 0.855 on Track 1, and F_{avg2} 0.940 with accuracy 0.922 on Track 2. The strongest official baselines reach 0.8425 on the Track 1 development set and 0.7475 on the Track 2 test set.

1 Introduction

Target-specific stance detection determines the viewpoint a text expresses toward a nominated target: FAVOR, AGAINST or NONE (Alturayeif et al., 2022). StanceEval-2026 (Albalawi et al., 2026a) poses the Arabic version over two tracks named for how far the evaluation target sits from the training data: *seen targets*, where the test target belongs to a thematic domain covered in training, and *unseen targets*, which tests cross-target generalization.

We took part in both tracks with one system.[1] Fine-tuned Arabic encoders, a few-shot classifier that retrieves labelled examples for a served instruction model, and generative models adapted with LoRA each give class probabilities. A weighted blend of these is turned into a label by a threshold rule with an explicit NONE abstention, whose thresholds are fixed on the development split.

We built the system one experiment at a time, each answering the question the previous one left open, and this paper reports that sequence, including eight controlled experiments that lowered F_{avg2}. The order shows which part of a few-shot pipeline carries transfer to a new target: the demonstrations, the labelled examples retrieved into the prompt, rather than the model. They had to be close to the test tweet in topic and surface form, and more of them did not help. Our official results are F_{avg2} 0.896 with accuracy 0.855 on Track 1 and F_{avg2} 0.940 with accuracy 0.922 on Track 2, against the strongest official baselines of F_{avg2} 0.8425 on the Track 1 development set and 0.7475 on the Track 2 test set.

Our contributions are: **(1)** the evidence for that finding, which places the bottleneck in the demonstration pool and in closeness rather than volume; **(2)** eight controlled experiments that lowered F_{avg2}, each reported against the configuration it replaced, one of which fails as the writing-style account predicts; **(3)** a decision rule that replaces argmax with per-class probability thresholds fixed on the development split, at half the attainable F_1 of each class.

2 Task, Data and Baselines

Data. Mawqif (Alturayeif et al., 2022) annotates Arabic tweets for stance, sentiment and sarcasm toward three targets. The 2026 task evaluates on Mawqif-XT (Albalawi et al., 2026b), 996 tweets on three new targets, Women Driving, E-Cars and the Trimester System, annotated under the same scheme by three annotators with a fourth resolving disagreements.

[1]Code, model checkpoints and generated data: https://huggingface.co/zaher-m/stanceeval2026

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Training uses the original 4,121-tweet corpus, split by the task into 3,502 train and 619 dev tweets over the seen targets. Track 1 tests on Women Driving (352 tweets); Track 2 trains on COVID-19 Vaccine and Digital Transformation only (2,721), holds out Women Empowerment (1,400) as an unseen-target development proxy, and tests jointly on E-Cars (332) and the Trimester System (312). Test labels are not available to participants.

Metric. F_{avg2} is the mean of the F_1 scores of FAVOR and AGAINST; F_{avg3} adds the F_1 of NONE. NONE has no F_1 term of its own in the main metric, but predicting it still removes the tweet from the other two counts, which Section 3.3 uses. NONE is rare, 7.1% of the Trimester tweets and 21.1% of E-Cars. Our scorer is checked against the official script from the task organizers.

Baselines. The organizers report fine-tuned AraBERT-twitter and MARBERT and zero-shot Qwen 2.5 (72B) and JAIS (70B), on the development set for Track 1 and on the test set for Track 2.

Related work. Training on one target and testing on another was identified early as the hard case of stance detection (Augenstein et al., 2016; Xu et al., 2018), and zero-shot stance detection extends it to targets never seen in training (Allaway and McKeown, 2020). In the 2024 edition of this task (Alturayeif et al., 2024) the winning system was an ensemble trained with contrastive losses and multi-task learning (Badran et al., 2024), and most teams fine-tuned Arabic encoders or sentence transformers (Melhem et al., 2024; Lichouri et al., 2024). Mawqif-XT turns fitting a target into transferring to one, where in-context learning enters: examples chosen for similarity to the query beat random ones (Liu et al., 2022; Rubin et al., 2022), and LLM-generated examples have been used to augment stance training data (Wagner et al., 2025). We use generation differently: not to train a model, but to fill the demonstration pool with on-target, surface-matched examples.

3 System

3.1 Components

Encoders. We fine-tune four Arabic encoders with a classification head on the (target, tweet) pair: MARBERTv2 (Abdul-Mageed et al., 2021), `bert-base-arabertv02-twitter` (Antoun et al., 2020), CAMeLBERT-mix (Inoue et al., 2021) and AraModernBERT (Elshehy et al., 2026). Track 2 encoders see a short target description before the tweet. Text is used as written (`preserve`) or with hashtags and elongation removed (`strip`).

Few-shot classifier. The classifier keeps a pool of labelled tweets, the demonstration pool. For each test tweet, a fine-tuned MARBERTv2 embeds the tweet (mean-pooled, ℓ_2-normalized) and retrieves class-balanced nearest neighbours from the pool, two per class; a served instruction model answers with one word given the six retrieved examples, sampled six times at temperature 0.7, and the votes form a soft distribution. The Arabic prompt asks for a direct answer with no intermediate reasoning. The primary backend is a 31B Gemma-4 instruction model in NVFP4; a 35B Qwen3.6 checkpoint was used in some runs.[2]

LoRA members. ALLaM-7B-Instruct (Bari et al., 2025) and Qwen2.5-7B-Instruct (Yang et al., 2024) are adapted with LoRA (Hu et al., 2022) at r=16, α=32 over all seven projection matrices, scored at inference by the log-probability of each label continuation. They are kept for the *direction* of their errors rather than their accuracy (Section 5.4).

Blend. Every component outputs a probability vector over the three classes: the encoders' softmax (averaged when several are used), the few-shot classifier's vote fractions, and a softmax over the LoRA models' label log-probabilities. The blend is a convex combination of these vectors, decoded by the rule of Section 3.3. On the development splits it puts 0.55 on the averaged encoders and 0.45 on the few-shot classifier, with a NONE offset of -0.1 on Track 1 and 0.0 on Track 2; weight and offset were selected by 5-fold stratified cross-validation on the development split, where cross-validated and full-split scores agree to within 0.001 (Table 4). On the blind test targets the encoders transferred poorly (Section 5.2), so the test-phase blend pairs the few-shot classifier with an ALLaM member instead, at equal weight on Track 1. The final Track 1 submission blends three members: the few-shot classifier over the style-matched pool at 0.427 and two ALLaM members trained on the Track 1 split at 0.309 and 0.264, set on the development split; Section 5.6 gives the Track 2 counterpart.

[2] `LilaRest/gemma-4-31B-it-NVFP4-turbo` and `nvidia/Qwen3.6-35B-A3B-NVFP4`, served through an OpenAI-compatible endpoint.

3.2 The Demonstration Pool

The pool starts as the training split, so at first it holds no tweets about the test target. We add generated tweets in two ways. In the first, a chat model writes tweets about the target in several dialects and tones; each is labelled by a separate classifier pass and kept only if that label agrees with the stance it was generated for. In the second, used for Women Driving, instead of describing the target we show the generator three real tweets from the test set as examples of writing style and seed the campaign's hashtags into the prompt. These test tweets carry no labels, since none were available to us, and none of them enters the pool. Only generated tweets do, filtered as above. The test set therefore supplies surface form (dialect, hashtags, letter elongation), not supervision. Adapting on unlabelled target-domain text is standard practice in domain adaptation (Gururangan et al., 2020).

3.3 Decision Rule

For one class c, $F_1 = 2\,\mathrm{TP}_c/(n_c + g_c)$, with n_c tweets predicted as c and g_c gold tweets of class c. Predicting c for one more tweet adds one to n_c and, with probability p (the model's probability for c), one to TP_c, so the expected change in F_1 is positive exactly when $p > F_1/2$. We therefore predict FAVOR or AGAINST when the class probability exceeds half the F_1 the classifier reaches for it, and NONE otherwise (about 0.45 with F_1 near 0.9). This is the standard threshold rule for F_1 (Lipton et al., 2014; Koyejo et al., 2014). An earlier form, used with the encoders, adds a fixed offset to the NONE probability before argmax. Both the class F_1 estimates and the offset are set on the development split before any test prediction.

4 Experimental Setup

All encoders share one training recipe (Appendix A). *Dev* scores are computed with our scorer on the development split, the in-domain 619 tweets for Track 1 and the held-out Women Empowerment tweets for Track 2. *Test* scores come from the official evaluation platform on the blind test sets; the two are different targets and are not compared.

5 Results

5.1 Which encoder, and does it matter

Table 1 gives each stage's score. We fine-tuned four encoder families with one recipe (Table 3). MARBERTv2 was best on both tracks, but its lead over AraBERT-twitter is 0.1 points on Track 1 dev, where the target is familiar, and 3.6 on Track 2 dev, where it is not; a gap that opens only when the target changes is a transfer problem, not a capacity one: the encoders rely on words tied to their training targets. Focal loss, class weighting and preprocessing changes did not help (Appendix B); combining encoders gave 0.847 (Track 1) and 0.840 (Track 2).

Stage	Set	Track 1	Track 2
Best single encoder	dev	0.8360	0.8195
Few-shot classifier, random examples	dev	0.8811	0.8879
+ retrieved examples	dev	0.8964	0.8942
Pool of training tweets only	test	0.815	0.851
+ generated on-target tweets	test	0.844	0.879
+ surface-matched tweets	test	0.853	0.870
Final system (Section 5.6): final pool, blend with LoRA members, decision rule of Section 3.3			
Official score of the final submission	test	0.896	0.940

Table 1: F_{avg2} at each stage; dev and test are different targets. Track 1 test rows are the equal-weight blend of the few-shot classifier and an ALLaM member; Track 2 test rows are the few-shot classifier with target descriptions.

5.2 The examples carry the transfer

A few-shot classifier with six random training examples scored 0.8811 on Track 1 dev and 0.8879 on Track 2 dev, beating the encoder combinations, by more on the unseen track. If examples carry the transfer, which ones reach the prompt should matter: retrieved examples, two per class, raised the scores to 0.8964 and 0.8942.

On the blind test the order held with larger gaps (Table 6, Appendix C): an encoder combination scored 0.694 on the Track 1 test target, the few-shot classifier 0.809 alone and 0.815 blended half and half with an ALLaM member; on Track 2 the classifier scored 0.826 alone and 0.851 with target descriptions, and a three-member blend with dev-tuned weights beat it on dev but lost on test.

The pool held only training tweets, so a new target retrieved nothing on-target. Adding generated on-target tweets (Section 3.2) was the largest gain: test F_{avg2} rose from 0.815 to 0.844 on Track 1 and from 0.851 to 0.879 on Track 2. Either the new examples are closer, or there are simply more of them. Doubling the generated pool and the examples per prompt made both tracks worse, 0.844 to 0.836 and 0.879 to 0.867: extra examples push the closest ones out of the prompt.

Change	ΔF_{avg2}	Set
Third blend member, different base	−0.0373	test
Reason step by step before answering	−0.0298	dev
Normalize hashtags and elongation	−0.0166	dev
Self-train on the model's own labels	−0.0159	dev
Stronger embedding model for retrieval	−0.0095	dev
Style matching on a target without a style	−0.0088	test
Real labelled tweets from another campaign	−0.0047	test
Retrain a blend member for accuracy	−0.0027	test

Table 2: Eight controlled experiments against the configuration each replaced; all lowered F_{avg2} and were dropped. Row 2 is chain-of-thought prompting (Wei et al., 2022); row 3 removes the surface features of Section 5.3; details in Appendix G.

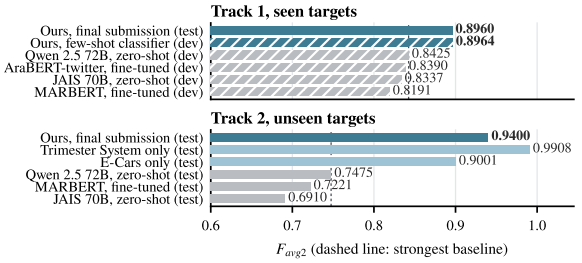


Figure 1: Official F_{avg2} of our final submissions (dark blue), our score on each Track 2 target (light blue) and the organizers' baselines (grey). Hatched bars are development-set scores, solid bars test-set scores.

5.3 Closeness includes writing style

Retrieval embeddings are sensitive to surface form, so matching the style of the test tweets should help beyond matching their topic. The Women Driving tweets are a clear case: dialectal, often sarcastic, and one campaign hashtag appears in 49% of them. With the pool generated in style mode (Section 3.2), test F_{avg2} rose from 0.8445 to 0.8530. The clearer evidence is retrieval reach: the share of test tweets whose nearest same-class example was on-target rose from 62% to 81% for FAVOR and from 72% to 87% for AGAINST (Appendix F).

The same change hurt Track 2 (0.8786 to 0.8698), as expected: E-Cars tweets have no campaign hashtag and no shared style. Style matching helps only where the target has a distinctive surface, and predicts that removing surface features should hurt (Section 5.4).

5.4 Eight experiments that lowered F_{avg2}

Table 2 lists eight further experiments; all lowered F_{avg2} against the configuration each replaced and were dropped. A third blend member on a different base was the largest loss: it disagreed with the blend on 29% of tweets but shared the few-shot classifier's AGAINST lean, so it added noise rather than balance; retraining the ALLaM member for accuracy lost because its FAVOR lean, which offset the classifier's, was its value.

The normalization row confirms the prediction of Section 5.3: splitting hashtags and collapsing repeated letters lowered the classifier from 0.8597 to 0.8431 on held-out real Women Driving tweets. The out-of-campaign row tests the same idea: 947 labelled Women Driving tweets from a later campaign with different hashtags scored below the style-matched generated pool on the test (Appendix K); real labels in the wrong style transferred worse than generated tweets in the right one.

5.5 Error analysis

On the Track 2 development proxy the reference classifier makes 149 errors on 1,400 tweets, 89 of them FAVOR–AGAINST confusions, almost all with a unanimous vote. Table 9 (Appendix J) shows the recurring kinds: complaints that empowerment is missing, read as opposition; irony and mock demands, read literally; and stance-free headlines and questions, read as taking a side.

5.6 Official results

The final submissions differ from the last stage rows of Table 1 in two ways. The blend: on Track 1 the equal-weight pair became the three members of Section 3.1; on Track 2, whose stage rows are the classifier alone, the ALLaM member trained on the Track 2 split was blended in the same way. The decoder: the rule of Section 3.3 replaced the NONE offset, with class thresholds estimated on the development split. Figure 1 compares the official scores with the organizers' baselines (all metrics in Appendix I). On Track 2, where baselines share our test set, ours is 19.3 points above the strongest. On Track 1 the baselines are on the development set, where our few-shot classifier reaches 0.8964 (+5.4).

6 Conclusion

The demonstration pool, not the model, decided transfer: examples close to the test tweet in topic and writing style carried it, and more examples hurt. Two of eight negative experiments became advice: blend members matter for the direction of their errors, and off-distribution validation sets mislead.

Limitations

The served models set the ceiling: with a 31B instruction model and 7B LoRA members, every model-side change we could afford lost accuracy (Section 5.4), and a stronger served model was out of reach on our hardware. Style matching needs a target with a distinctive surface; on E-Cars it hurt. The decision rule optimizes F_{avg2} and underpredicts NONE by design, so F_{avg3} trails F_{avg2} (Appendix I). Development data from other targets misled us twice (Section 5.3, Appendix K), so decisions that depend on a new target's style cannot be validated before test. Generated demonstrations were filtered by a classifier, not checked by annotators, and we measured nothing outside dialectal Arabic tweets. Every score is a single run of a sampled classifier; on the development proxy two runs of one configuration differed by 0.012 (Appendix F), so differences below about 0.01 in Tables 1 and 2 are within run-to-run variation.

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References

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088–7105, Online. Association for Computational Linguistics.

Rasha Albalawi, Nuha Albadi, Hamzah Luqman, Saad Ezzini, Maram Kurdi, Asma Yamani, Ahmed Ashraf, Maged Al-Shaibani, and Nora Alturayeif. 2026a. StanceEval-2026: The second stance detection shared task. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Rasha Albalawi, Nuha Albadi, Hamzah Luqman, Maram Kurdi, Saad Ezzini, Asma Yamani, and Ahmed Ashraf. 2026b. Mawqif-XT: An Arabic benchmark dataset for cross-target stance detection. *Preprint*, arXiv:2608.09539.

Ali Alkhathlan, Faris Alahmadi, Faris Kateb, and Hend Al-Khalifa. 2025. Constructing and evaluating ArabicStanceX: a social media dataset for Arabic stance detection. *Frontiers in Artificial Intelligence*, 8:1615800.

Emily Allaway and Kathleen McKeown. 2020. Zero-shot stance detection: A dataset and model using generalized topic representations. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8913–8931, Online. Association for Computational Linguistics.

Nora Alturayeif, Hamzah Luqman, Zaid Alyafeai, and Asma Yamani. 2024. StanceEval 2024: The first Arabic stance detection shared task. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 774–782, Bangkok, Thailand. Association for Computational Linguistics.

Nora Saleh Alturayeif, Hamzah Abdullah Luqman, and Moataz Aly Kamaleldin Ahmed. 2022. Mawqif: A multi-label Arabic dataset for target-specific stance detection. In *Proceedings of the Seventh Arabic Natural Language Processing Workshop (WANLP)*, pages 174–184, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. AraBERT: Transformer-based model for Arabic language understanding. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection*, pages 9–15, Marseille, France. European Language Resource Association.

Isabelle Augenstein, Tim Rocktäschel, Andreas Vlachos, and Kalina Bontcheva. 2016. Stance detection with bidirectional conditional encoding. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 876–885, Austin, Texas. Association for Computational Linguistics.

Mohamed Badran, Mo'men Hamdy, Marwan Torki, and Nagwa El-Makky. 2024. AlexUNLP-BH at StanceEval2024: Multiple contrastive losses ensemble strategy with multi-task learning for stance detection in Arabic. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 823–827, Bangkok, Thailand. Association for Computational Linguistics.

M Saiful Bari, Yazeed Alnumay, Norah A. Alzahrani, Nouf M. Alotaibi, Hisham A. Alyahya, Sultan AlRashed, Faisal A. Mirza, Shaykhah Z. Alsubaie, Hassan A. Alahmed, Ghadah Alabduljabbar, Raghad Alkhathran, Yousef Almushayqih, Raneem Alnajim, Salman Alsubaihi, Maryam Al Mansour, Majed Alrubaian, Ali Alammari, Zaki Alawami, Abdulmohsen Al-Thubaity, and 6 others. 2025. ALLaM: Large language models for Arabic and English. In *The Thirteenth International Conference on Learning Representations (ICLR 2025)*. OpenReview.net.

Omar Elshehy, Omer Nacar, Abdelbasset Djamai, Muhammed Ragab, Khloud Al Jallad, and Mona Abdelazim. 2026. AraModernBERT: Transtokenized

initialization and long-context encoder modeling for Arabic. In *Proceedings of the 2nd Workshop on NLP for Languages Using Arabic Script*, pages 313–321, Rabat, Morocco. Association for Computational Linguistics.

Suchin Gururangan, Ana Marasović, Swabha Swayamdipta, Kyle Lo, Iz Beltagy, Doug Downey, and Noah A. Smith. 2020. Don't stop pretraining: Adapt language models to domains and tasks. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8342–8360, Online. Association for Computational Linguistics.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *The Tenth International Conference on Learning Representations (ICLR 2022)*, Virtual Event. OpenReview.net.

Go Inoue, Bashar Alhafni, Nurpeiis Baimukan, Houda Bouamor, and Nizar Habash. 2021. The interplay of variant, size, and task type in Arabic pre-trained language models. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop (WANLP)*, pages 92–104, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Oluwasanmi O. Koyejo, Nagarajan Natarajan, Pradeep K. Ravikumar, and Inderjit S. Dhillon. 2014. Consistent binary classification with generalized performance metrics. In *Advances in Neural Information Processing Systems 27 (NIPS 2014)*. Curran Associates, Inc.

Mohamed Lichouri, Khaled Lounnas, Ouaras Rafik, Mohamed ABi, and Anis Guechtouli. 2024. dzStance at StanceEval2024: Arabic stance detection based on sentence transformers. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 794–799, Bangkok, Thailand. Association for Computational Linguistics.

Zachary C. Lipton, Charles Elkan, and Balakrishnan Naryanaswamy. 2014. Optimal thresholding of classifiers to maximize F1 measure. In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2014)*, pages 225–239. Springer.

Jiachang Liu, Dinghan Shen, Yizhe Zhang, Bill Dolan, Lawrence Carin, and Weizhu Chen. 2022. What makes good in-context examples for GPT-3? In *Proceedings of Deep Learning Inside Out (DeeLIO 2022): The 3rd Workshop on Knowledge Extraction and Integration for Deep Learning Architectures*, pages 100–114, Dublin, Ireland and Online. Association for Computational Linguistics.

Anas Melhem, Osama Hamed, and Thaer Sammar. 2024. TAO at StanceEval2024 shared task: Arabic stance detection using AraBERT. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 842–846, Bangkok, Thailand. Association for Computational Linguistics.

Ohad Rubin, Jonathan Herzig, and Jonathan Berant. 2022. Learning to retrieve prompts for in-context learning. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 2655–2671, Seattle, United States. Association for Computational Linguistics.

Stefan Sylvius Wagner, Maike Behrendt, Marc Ziegele, and Stefan Harmeling. 2025. The power of LLM-generated synthetic data for stance detection in online political discussions. In *The Thirteenth International Conference on Learning Representations (ICLR 2025)*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, pages 24824–24837.

Chang Xu, Cécile Paris, Surya Nepal, and Ross Sparks. 2018. Cross-target stance classification with self-attention networks. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 778–783, Melbourne, Australia. Association for Computational Linguistics.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, and 23 others. 2024. Qwen2.5 technical report. *Preprint*, arXiv:2412.15115.

A The Experiment Sequence

Figure 2 lays out the eight experiments in order and Figure 3 the system they produced. Each experiment below is reported with the same fields: the question, the data and processing, the models and settings, the result, and what it ruled out. Scores marked dev are computed with our scorer on the development split (Track 1: the in-domain 619-tweet split; Track 2: the held-out Women Empowerment tweets). Scores marked test come from the official evaluation platform on the blind test sets.

Common settings. All encoders: maximum length 128, batch size 16, evaluation batch 64, learning rate 2×10^{-5}, weight decay 0.01, cross-entropy unless stated, 12 epochs with patience 3, seed 42, label order `[Against, Favor, None]`. Few-shot classifier: 6 examples, 2 per class; 6 samples at temperature 0.7; at most 8 new tokens; Arabic prompt asking for a direct one-word answer; retriever MARBERTv2 with mean pooling

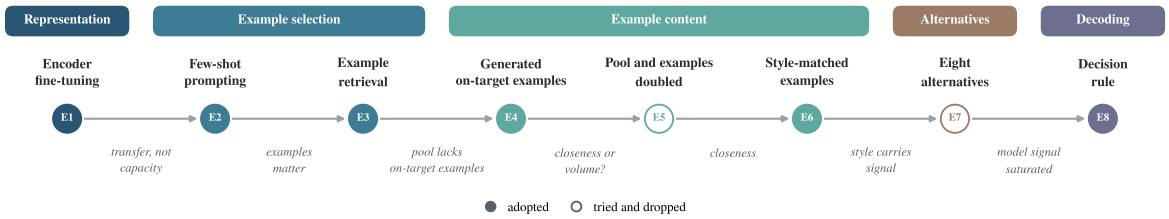


Figure 2: The eight experiments in the order they ran. The label under each arrow is the finding that led to the next step. Filled markers were adopted; hollow markers were tried and dropped.

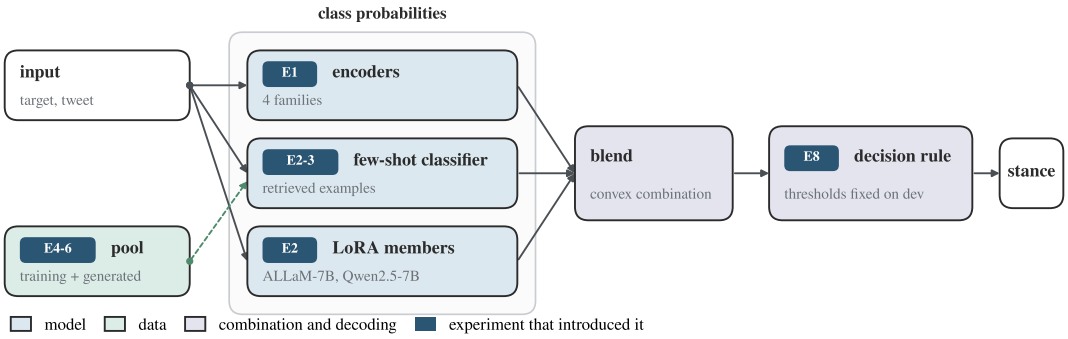


Figure 3: The system, with each component tagged by the experiment that introduced it: E1 encoders, E2 few-shot prompting and LoRA members, E3 retrieval, E4 to E6 the demonstration pool, E8 the decision rule. E7 added nothing: all eight of its experiments were dropped.

and unit-length vectors. LoRA: rank 16, α 32, on `q`, `k`, `v`, `o`, `gate`, `up` and `down`; label probability from the log-probability of the label word. Example generation: 4 dialects $\times$ 5 tones; in style mode, 3 real unlabelled test tweets as examples and the campaign hashtags in the prompt; a generated tweet is admitted only if a separate classifier pass agrees with the label it was generated for.

B E1: Encoders

Question. Which encoder, and is the encoder the bottleneck?

Data and processing. Track 1: 3,502 training and 619 development tweets over the three seen targets. Track 2: 2,721 training tweets (COVID-19 Vaccine, Digital Transformation) and the 1,400 Women Empowerment tweets as development data. Input is the target and the tweet; Track 2 encoders prepend a short description of the target. Two text forms: `preserve` keeps hashtags and letter elongation, `strip` removes them.

Models and settings. Four bases under the common recipe: MARBERTv2, AraBERTv02-twitter, CAMeLBERT-mix, AraModernBERT. Loss variants on MARBERTv2 (Track 2): class-weighted cross-entropy, focal loss with γ=2. A paraphrase-augmented training split (Appendix L) was also tried.

Result. Table 3 lists all fourteen released checkpoints with their dev F_{avg2} at the best epoch. MARBERTv2 leads AraBERT-twitter by 0.1 points on Track 1 (0.8360 against 0.8351) and by 3.6 on Track 2 (0.8195 against 0.7837). On Track 2 both loss changes lose against plain cross-entropy: class-weighted 0.8088 (-0.011) and focal 0.7794 (-0.040). Preprocessing is model-dependent: `strip` helps AraBERT-twitter ($+0.003$) and hurts MARBERTv2 (-0.005). The augmented split hurts a single model (0.7930, -0.027) but helps as an added member. Combining two or three encoders reaches 0.8465 on Track 1 dev and 0.8403 on Track 2 dev (Table 4), with the NONE offset chosen by 5-fold cross-validation; the cross-validated and full-split scores agree to within 0.001 in every row.

What it ruled out. Encoder choice and loss design as the lever. The gap that opens only when the target changes is a transfer problem.

Checkpoint	Base	Dev	Ep.
Track 1 (in-domain dev)			
`t1_arabert_strip`	AraBERT-tw	0.8385	7
`t1_marbert`	MARBERTv2	0.8360	3
`t1_arabert`	AraBERT-tw	0.8351	2
`t1_marbert_strip`	MARBERTv2	0.8311	6
`t1_camelbert`	CAMeLBERT-mix	0.7904	3
`t1_aramodern`	AraModernBERT	0.7897	10
Track 2 (held-out Women Empowerment)			
`t2_marbert`	MARBERTv2	0.8195	2
`t2_marbert_strip`	MARBERTv2	0.8147	3
`t2_marbert_weighted`	MARBERTv2	0.8088	3
`t2_marbert_aug`	MARBERTv2	0.7930	3
`t2_arabert`	AraBERT-tw	0.7837	3
`t2_marbert_focal`	MARBERTv2	0.7794	3
`t2_camelbert`	CAMeLBERT-mix	0.7506	2
`t2_aramodern`	AraModernBERT	0.7308	5

Table 3: The 14 released encoders, sorted by dev F_{avg2} within track. Ep. is the epoch of the best score. `t1_marbert` and `t2_marbert` also serve as the retrieval embedders.

Track	Members	Offset	CV	Dev
1	AraBERT + MARBERTv2	+0.1	0.8384	0.8392
1	+ AraBERT(strip)	−0.2	0.8469	0.8465
2	AraBERT + MARBERTv2	−0.6	0.8300	0.8300
2	+ MARBERTv2(strip)	−0.6	0.8310	0.8313
2	+ MARBERTv2(aug)	−0.2	0.8402	0.8403
1	0.55 encoders + 0.45 few-shot	−0.1	0.8953	0.8949
2	0.55 encoders + 0.45 few-shot	0.0	0.8892	0.8891

Table 4: Encoder combinations and encoder–few-shot blends on dev F_{avg2}. Offset is the NONE-probability offset chosen by 5-fold stratified cross-validation; CV and full-dev scores agree throughout.

C E2: Few-Shot Classifier and LoRA Members

Question. Does a model that is told the task transfer better than one trained on the targets?

Data and processing. Examples drawn from the training split, two per class. LoRA members fine-tuned on the Track 2 training split (COVID-19 Vaccine, Digital Transformation) and evaluated on the held-out Women Empowerment tweets.

Models and settings. Few-shot classifier as in the common settings, with six random examples. LoRA on ALLaM-7B-Instruct and Qwen2.5-7B-Instruct. Seven adapters are released (Table 5).

Result. On dev the few-shot classifier scored 0.8811 (Track 1) and 0.8879 (Track 2), against 0.847 and 0.840 for the best encoder combinations. Fine-tuned generative members transfer to the unseen target far better than encoders do (ALLaM 0.883, Qwen2.5 0.862, against 0.69 for an encoder on the same held-out target) but trail the few-shot classifier. On the blind test (Table 6) the same order holds with larger gaps, and a three-member blend with weights tuned on dev beat the classifier alone on dev and lost to it on test.

Adapter	Base	Training data
`allam_t2`	ALLaM-7B	Track 2 split
`allam_full`	ALLaM-7B	Track 1 split
`allam_v2`	ALLaM-7B	Track 1 split
`allam_loo`	ALLaM-7B	Track 1 split, one target held out
`allam_real`	ALLaM-7B	real tweets, training targets
`allam_style`	ALLaM-7B	style-matched generated set
`qwen_t2`	Qwen2.5-7B	Track 2 split

Table 5: The seven released LoRA adapters. `allam_t2` is the Track 2 blend member; `allam_full` and `allam_v2` are the Track 1 blend members (weights 0.309 and 0.264 in the final submission); `allam_real` and `allam_style` are the two retrained members of E7; `qwen_t2` is the third member tried in E7.

Track	Configuration	Test
1	encoder combination only	0.694
1	0.55 encoders + 0.45 few-shot	0.787
1	few-shot + target descriptions	0.798
1	few-shot, retrieved examples	0.809
1	0.5 few-shot + 0.5 ALLaM	**0.815**
2	ALLaM fine-tuned	0.804
2	ALLaM + Qwen blend	0.810
2	three-member blend, weights tuned on dev	0.817
2	few-shot, retrieved examples	0.826
2	few-shot + target descriptions	**0.851**

Table 6: Component families on the blind test sets, before the demonstration pool was changed. Bold rows are the starting points of the test rows in Table 1.

What it ruled out. Fine-tuning on the training targets as the main predictor. Dev-tuned blend weights as a reliable choice for a different test target.

D E3: Retrieval

Question. If examples carry the transfer, does the choice of examples matter?

Data and processing. Same pool as E2. Retrieval by cosine similarity between the fine-tuned MARBERTv2 embedding of the test tweet and of each pool tweet, keeping the nearest two per class.

Models and settings. Common few-shot settings. Alternatives tried for retrieval: sentence-

embedding models built for Arabic semantic similarity (Arabic-Triplet-Matryoshka-V2, bge-m3) and a reranker (GATE), and, later, Qwen3-Embedding-4B (E7).

Result. On dev, retrieved examples beat random ones: 0.8811 to 0.8964 on Track 1 and 0.8879 to 0.8942 on Track 2. Six examples was the best count. The semantic-similarity embedders and the reranker scored below MARBERTv2 as retrievers; similarity for search is not the same objective as picking useful stance examples for dialectal tweets.

What it ruled out. Random example selection; off-the-shelf semantic retrievers for this task.

E E4 and E5: Generated Examples and Their Number

Question. The pool held no tweets about the test targets. Does adding generated on-target tweets help, and if so, because they are close or because there are more of them?

Data and processing. For each test target, a chat model wrote tweets varying dialect (Saudi, Egyptian, Levantine, Modern Standard) and tone (serious, sarcastic, question, emotional, casual). Each generated tweet was labelled by a separate classifier pass and kept only if the label agreed with the stance it was generated for. About 400 kept tweets per target were added to the pool.

Models and settings. Common few-shot settings, six examples. E5 doubled both the pool and the number of examples per prompt.

Result. On the blind test, E4 moved F_{avg2} from 0.815 to 0.844 on Track 1 and from 0.851 to 0.879 on Track 2, with the Trimester System moving from 0.841 to 0.904. E5, doubling, made both tracks worse: 0.844 to 0.836 and 0.879 to 0.867.

What it ruled out. Volume as the explanation. The gain comes from closeness; extra examples push the closest ones out of the prompt.

F E6: Style Matching

Question. Does matching the writing style of the test tweets help beyond matching their topic?

Data and processing. Generation prompted with three real unlabelled test tweets as examples of style and the campaign hashtags seeded in. No test labels exist for participants and none were used; no test tweet entered the pool. Generated tweets were filtered as in E4. The released style-matched set has 3,902 rows (Appendix L).

Track 1 configuration	Test
generated examples + ALLaM	0.8445
0.25 style-matched + ALLaM	0.8467
0.5 style-matched + ALLaM, 41 NONE	0.8427
0.5 style-matched + ALLaM, 25 NONE	0.8499
0.6 style-matched + ALLaM, 18 NONE	0.8499
0.5 style-matched + ALLaM, 18 NONE	**0.8530**
ALLaM retrained on the generated pool	0.8503
real tweets from another campaign, alone	0.8434
real tweets from another campaign + ALLaM	0.8483
real + generated pool, ALLaM retrained on real	0.8436

Table 7: E6 and two E7 rows on Track 1 test. The upper block lists the submitted style-matched configurations, which differ in the style-matched weight and the NONE count; the lower block replaces generated examples with real labelled tweets from a different campaign, or retrains the blend member. All four lower rows lose.

Models and settings. Common few-shot settings; blend 0.5 few-shot + 0.5 ALLaM on Track 1; 18 NONE predictions.

Result. Track 1 test F_{avg2} 0.8445 to 0.8530 (Table 7). Retrieval reach, the share of test tweets whose nearest same-class example was on-target rather than about a training target, rose from 62% to 81% for FAVOR and from 72% to 87% for AGAINST. Submissions with 25 and 41 NONE predictions scored lower. On Track 2 the same recipe fell from 0.8786 to 0.8698 (E-Cars 0.8596 to 0.8479, Trimester System 0.9036 to 0.9000) although it had looked good on the development proxy (0.9021 to 0.9143 over two runs).

What it ruled out. Style matching as a general recipe. It helps where the target has a distinctive surface and hurts where it does not, and the development proxy did not predict which.

G E7: Eight Controlled Experiments That Lowered F_{avg2}

Question. With a stronger served model out of reach, can anything on the model side still add accuracy?

Data and processing. Dev experiments use the Track 2 development proxy at its best configuration (direct answering with target descriptions, 0.9049). Test experiments use the Track 1 blend at 0.8530 or the Track 2 configuration at 0.8786. Every change

is compared against the configuration it replaced (Table 2 in the body).

Result. All eight lost. Chain-of-thought prompting fell to 0.8751; on the 83 tweets where it disagreed with direct answering, direct was right 68.7% of the time and the reasoning chain 19.3%, and FAVOR-to-AGAINST errors rose from 61 to 91. A stronger retriever (Qwen3-Embedding-4B) tied MARBERTv2 as a nearest-neighbour stance predictor (0.666 against 0.665 at k=6) and lost end to end at 0.8954, disagreeing on 3.5% of tweets and being wrong on 73% of those. Self-training on the model's own predictions fell to 0.889, and consensus labels from several models, 94% accurate, fell further to 0.884; the failure is self-confirmation, not label noise. Ensembling three sampled example sets fell to 0.898. A third blend member on Qwen2.5-7B cost more the more weight it got: 0.8530 to 0.8250 at weight 0.15 and to 0.8157 at 0.25. Normalizing hashtags and elongation fell from 0.8597 to 0.8431 on a real held-out set. A diagnostic explains why so little moved: the six-sample vote was unanimous on 99.6% of tweets, so the model was confidently wrong on its 149 errors, 89 of them FAVOR–AGAINST confusions, and NONE was already at 24 predictions, so no threshold change helped.

What it ruled out. Reasoning prompts, better retrievers, self-training, decorrelated members and surface normalization, under the served models available to us. A separate negative concerns validation: a generated development set for an unseen target, built to tune decisions where no gold exists, tracked realized scores in level (Pearson 0.94) but not in order (Spearman at most 0.4), so it was used only for coarse checks.

H E8: Decision Rule

Section 3.3 gives the rule: predict FAVOR or AGAINST when the class probability exceeds half the class F_1, and NONE otherwise. Class F_1 is estimated on the development split (about 0.9, giving a threshold near 0.45). The earlier form, an additive offset on the NONE probability, is chosen by 5-fold stratified cross-validation on dev: -0.1 for the Track 1 blend and 0.0 for Track 2 (Table 4), where it was worth up to a point of dev F_{avg2}. All parameters are fixed on dev before test prediction.

Block	N	F_{avg2}	F_{avg3}	Acc.
Track 1: Women Driving	352	0.8960	0.6862	0.8551
Track 2: E-Cars	332	0.9001	0.8310	0.8584
Track 2: Trimester System	312	0.9908	0.9780	0.9904
Track 2: overall	644	0.9400	0.8792	0.9224

Table 8: Official metrics of our final submissions, per target block. Track 2 is scored over the concatenation of its two blocks, so the overall row is not the mean of the two above it.

I Official Results

Table 8 gives the organizers' three metrics for our final submission on each target block; the organizers' baselines are in Figure 1. The two Track 2 targets differ by 9.1 points of F_{avg2}, with the Trimester System, whose tweets share a campaign style, close to solved. F_{avg3} trails F_{avg2} on both tracks because the decision rule under-predicts NONE by design; where NONE matters, re-fit the threshold.

J Error Examples

Table 9 gives six errors of the reference few-shot classifier (direct answering with target descriptions) on the Track 2 development proxy, Women Empowerment, chosen from the three confusion types that account for nearly all its errors. Emojis, links and mentions are removed.

K Real Tweets From Another Campaign

The strongest test of the style-matching explanation replaces generated examples with real ones. We took 947 labelled Women Driving tweets from ArabicStanceX (Alkhathlan et al., 2025), a public corpus covering a different, later campaign, and checked that none of them appears in the test set. On a validation split from that corpus the real examples looked decisive: 0.8597 against 0.8141 for generated examples and 0.7644 with no on-target examples at all. On the blind test they scored below the style-matched generated pool at every blend weight (Table 7). The two campaigns differ in hashtags and era, so the real tweets sit in the wrong region of surface form: matching the test set's surface with generated text beat matching its target with real text, and a validation set from a different distribution pointed the wrong way.

L Released Data and Artifacts

Everything is on Hugging Face (footnote 1): `models/encoders/` holds the 14 encoders with the

Tweet	Gloss	Gold	Predicted	Reading
وين تمكين المرأة وش هالكلام	Where is women's empowerment? What is this talk?	Favor	Against	complaint that the target is missing, read as opposition
يا قلبي على تمكين المرأة.. بكى والله	My heart goes out to women's empowerment.. it wept, I swear	Favor	Against	lament over the target's state, read as opposition
نطالب بقمع الرجال تمكين المرأة لايكفي	We demand that men be suppressed; women's empowerment is not enough	Against	Favor	mock demand, read literally
أمةً انجازها حجاب فاشلة أمة انجازها تمكين المرأة ناجحة	A nation whose achievement is the hijab is a failure; a nation whose achievement is women's empowerment is a success	Against	Favor	ironic restatement of the opposing view, read as the writer's own
جامعة الشرق الأوسط تنظم ملتقى تمكين المرأة الاقتصادي	Middle East University organizes a forum on women's economic empowerment	None	Favor	headline with no stance, read as support
هنا نتذكر مقولة تمكين المرأة ! ماهي المعايير التي قامت عليها هذه المقوله ؟	Here we recall the slogan "women's empowerment"! On what criteria was this slogan built?	None	Against	question about the slogan, read as opposition

Table 9: Six errors of the reference few-shot classifier on the Women Empowerment development proxy, one or two per confusion type. Gold labels are the organizers' annotations; the classifier's six-sample vote was unanimous on every row.

`best.json` each was trained under, `models/lora/` the 7 adapters with their `adapter_config.json`, `code/` and `configs/` the pipeline, and `data/` three generated or augmented training sets with rows `(text, target, stance)`: `track2/train_aug` (8,056 rows: the Track 2 training split plus two paraphrases per tweet, marked in a `source` column; trains `t2_marbert_aug`), `synth/t1_train_style` (3,902 rows: the Track 1 training split plus the style-matched generated tweets; trains `allam_style`) and `external/pool_real_all` (4,449 rows of pooled real tweets on the training targets; trains `allam_real`). The organizers' splits are not redistributed; the released code regenerates every set here from them.